

$$T : V \longrightarrow W$$

$$\downarrow \mathcal{B}_V, \mathcal{B}_W$$

$$A = [T]_{\mathcal{B}_V, \mathcal{B}_W}$$

$$T^* : W^* \longrightarrow V^*$$

$$\downarrow \mathcal{B}_W^*, \mathcal{B}_V^*$$

$$A^t = [T^*]_{\mathcal{B}_W^*, \mathcal{B}_V^*}$$